Complete Stock Market Parameters & Formulas Reference Guide

For AI-Powered Indian Stock Prediction System

PART 1: TECHNICAL ANALYSIS INDICATORS

1.1 RELATIVE STRENGTH INDEX (RSI)

Definition: Momentum oscillator measuring magnitude of recent price changes to evaluate overbought/oversold conditions (0-100 scale)

Formula:

```
RSI = 100 - [100 / (1 + RS)]

Where:
RS = Average Gain / Average Loss

Average Gain = Sum of gains over N periods / N
Average Loss = Sum of losses over N periods / N

Wilder's Smoothing (for periods after initial 14):
Avg_Gain_n = [(Prev_Avg_Gain × 13) + Current_Gain] / 14
Avg_Loss_n = [(Prev_Avg_Loss × 13) + Current_Loss] / 14
```

Parameters:

- Standard Period: 14 trading days
- Overbought Level: RSI > 70 (sell signal)
- Oversold Level: RSI < 30 (buy signal)
- Neutral Zone: 40-60

Interpretation:

- RSI > 70: Stock likely overvalued, potential pullback
- RSI < 30: Stock likely undervalued, potential bounce
- Divergence: Price makes new high but RSI doesn't (bearish)

1.2 MOVING AVERAGE CONVERGENCE DIVERGENCE (MACD)

Definition: Trend-following momentum indicator showing relationship between two moving averages

Formula:

```
MACD_Line = EMA_12 - EMA_26

Signal_Line = EMA_9(MACD_Line)

Histogram = MACD_Line - Signal_Line

Where EMA_n = (Close × Multiplier) + (EMA_prev × (1 - Multiplier))
      Multiplier = 2 / (n + 1)

EMA_12 Multiplier = 2/13 ≈ 0.1538
EMA_26 Multiplier = 2/27 ≈ 0.0741
Signal Line Multiplier = 2/10 = 0.2
```

Parameters:

- Fast EMA: 12 periods
- Slow EMA: 26 periods
- Signal EMA: 9 periods

Interpretation:

- MACD > Signal: Bullish (buy signal), upward momentum
- MACD < Signal: Bearish (sell signal), downward momentum

- Histogram > 0: MACD above signal, uptrend strength
- Histogram < 0: MACD below signal, downtrend strength
- Zero-line crossover: Trend change confirmation

1.3 BOLLINGER BANDS

Definition: Volatility bands placed above/below moving average; bands widen during high volatility, narrow during low volatility

Formula:

```
Middle Band (SMA_20) = Σ(Close_i) / 20, where i = last 20 periods

Standard Deviation = √[Σ(Close_i - SMA_20)² / 20]

Upper Band = SMA_20 + (2 × Std Dev)
Lower Band = SMA_20 - (2 × Std Dev)

%B = (Close - Lower_Band) / (Upper_Band - Lower_Band)
Where: %B = 0 at lower band, %B = 1 at upper band, %B = 0.5 at middle
```

Parameters:

- Period: 20 trading days
- Standard Deviation Multiplier: 2 (covers ~95% of price movements)

Interpretation:

- Price > Upper Band: Overbought (potential reversal)
- Price < Lower Band: Oversold (potential reversal)
- Band Width: Volatility indicator (narrow = low volatility, wide = high volatility)
- %B > 0.9: At upper band, strong overbought
- %B < 0.1: At lower band, strong oversold
- Band Squeeze: Low volatility, breakout expected

1.4 ON-BALANCE VOLUME (OBV)

Definition: Cumulative indicator using volume and price direction to measure buying/selling pressure

Formula:

```
If Close_t > Close_(t-1):
   OBV_t = OBV_(t-1) + Volume_t

If Close_t < Close_(t-1):
   OBV_t = OBV_(t-1) - Volume_t

If Close_t = Close_(t-1):
   OBV_t = OBV_(t-1)

OBV_Trend = SMA_20(OBV) = Σ(OBV_i) / 20
```

Interpretation:

- OBV Rising with Price: Strong buying pressure (bullish)
- OBV Falling with Price: Strong selling pressure (bearish)
- OBV Divergence: OBV up but price down = potential upside (bullish)
- Volume Confirmation: High OBV in direction of price = strong move

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1.5 STOCHASTIC OSCILLATOR

Definition: Momentum indicator comparing closing price to price range over period; ranges 0-100

Formula:

```
%K = [(Close - Lowest Low over N) / (Highest High over N - Lowest Low over N)] × 100

%D = SMA_3(%K) = Simple Moving Average of %K over 3 periods

Where N = 14 periods (standard)
```

Parameters:

- Period: 14 trading days
- Smoothing: 3-period SMA for %D
- Overbought: > 80
- Oversold: < 20

1.6 AVERAGE TRUE RANGE (ATR)

Definition: Volatility indicator measuring average price range over period; NOT directional

Formula:

```
True Range (TR) = max(
    High - Low,
    |High - Close_prev|,
    |Low - Close_prev|
)

ATR = SMA_14(TR) = Σ(True_Range_i) / 14

Where N = 14 periods (standard)
```

Interpretation:

- High ATR: High volatility, wider stop-losses recommended

- Low ATR: Low volatility, tighter stop-losses acceptable
- ATR Expansion: Breakout likely
- ATR Contraction: Range-bound, consolidation phase

1.7 AVERAGE DIRECTIONAL INDEX (ADX)

Definition: Strength of trend indicator (NOT directional); measures whether trend exists regardless of direction

Formula:

```
+DM = max(High_t - High_(t-1), 0)
-DM = max(Low_(t-1) - Low_t, 0)

If High_t - High_(t-1) > Low_(t-1) - Low_t: -DM = 0
If Low_(t-1) - Low_t > High_t - High_(t-1): +DM = 0

+DI = (+DM / ATR) × 100
-DI = (-DM / ATR) × 100

DX = |+DI - -DI| / (+DI + -DI) × 100

ADX = SMA_14(DX)
```

Parameters:

- Period: 14 trading days
- ADX < 25: Weak trend
- ADX 25-50: Moderate trend strength
- ADX > 50: Strong trend

PART 2: FINANCIAL METRICS & FUNDAMENTAL ANALYSIS

2.1 EARNINGS PER SHARE (EPS)

Definition: Company's profit attributable to each outstanding share; key profitability metric

Formula:

```
EPS = (Net Income - Preferred Dividends) / Weighted Average Shares Outstanding

Trailing EPS (TTM) = Sum of last 4 quarters earnings / Weighted Avg Shares

Forward EPS = Projected earnings next 12 months / Current shares outstanding
```

Interpretation:

- Higher EPS: More profitable per share
- EPS Growth: Quarter-over-quarter or year-over-year increase
- Negative EPS: Company operating at loss

Related Metrics:

- Diluted EPS: Includes potential share dilution from options/warrants
- Basic EPS: Simple calculation without dilution

2.2 PRICE-TO-EARNINGS RATIO (P/E)

Definition: Valuation metric showing price paid per unit of earnings; lower = potentially cheaper

Formula:

```
P/E Ratio = Current Stock Price / Earnings Per Share (EPS)

Also expressed as:
P/E = Market Capitalization / Total Net Income

Justified P/E = Dividend Payout Ratio / (Required Return - Growth Rate)
            = D/P / (R - G)

Where:
D/P = Annual Dividend / Current Price
R = Required Rate of Return
G = Sustainable Growth Rate
```

Interpretation:

- P/E < 15: Potentially undervalued (depending on sector)
- P/E 15-25: Fair valuation
- P/E > 25: Potentially overvalued or high-growth stock
- Trailing P/E: Based on past 12 months earnings
- Forward P/E: Based on estimated next 12 months earnings

Comparison:

- Compare to industry average P/E
- Compare to historical average P/E of same stock
- Context: Growth stage, profitability, sector norms

2.3 PRICE-TO-BOOK RATIO (P/B)

Definition: Valuation metric comparing market price to book value per share

Formula:

```
P/B Ratio = Current Stock Price Per Share / Book Value Per Share

Book Value Per Share = (Total Assets - Total Liabilities) / Shares Outstanding

Also: P/B = Market Capitalization / Total Shareholders' Equity
```

Interpretation:

- P/B < 1.0: Stock trading below book value (potential value opportunity)
- P/B > 1.0: Stock trading above book value (market valuing intangibles)
- P/B 1-3: Normal range for healthy companies
- High P/B: Growth expectations or intangible assets (brand, patents)

2.4 DIVIDEND YIELD

Definition: Annual dividend income relative to stock price; income measure for dividend stocks

Formula:

```
Dividend Yield = (Annual Dividend Per Share / Current Stock Price) × 100%

Also expressed as:
Dividend Yield = (Total Dividends / Market Cap) × 100%

Payout Ratio = Dividends Per Share / Earnings Per Share × 100%
```

Interpretation:

- Dividend Yield > 5%: High yield, attractive for income (verify sustainability)
- Dividend Yield 2-5%: Moderate, common for mature companies
- Dividend Yield < 2%: Low yield, focus on growth

- Rising Yield: Either higher dividend or falling stock price
- Payout Ratio > 50%: Company distributing more than half earnings

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2.5 DEBT-TO-EQUITY RATIO

Definition: Leverage metric showing financial risk from debt usage

Formula:

```
D/E Ratio = Total Liabilities / Total Shareholders' Equity

Also expressed as:
D/E = Total Debt / Total Equity

Interpretation:
D/E = 1.0: Equal debt and equity
D/E < 1.0: More equity than debt (lower risk)
D/E > 1.0: More debt than equity (higher risk)
```

Interpretation:

- D/E < 0.5: Conservative, low financial risk
- D/E 0.5-1.5: Moderate leverage, acceptable
- D/E > 2.0: High leverage, higher bankruptcy risk
- Compare to industry average for context

2.6 RETURN ON EQUITY (ROE)

Definition: Profitability metric measuring how efficiently company uses shareholder capital

Formula:

```
ROE = Net Income / Average Shareholders' Equity × 100%

ROE = Net Income / (Beginning Equity + Ending Equity) / 2 × 100%
```

Interpretation:

- ROE > 15%: Excellent management, good returns on capital
- ROE 10-15%: Good, above average
- ROE 5-10%: Moderate
- ROE < 5%: Poor capital efficiency
- Increasing ROE: Improving profitability or reducing equity base

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2.7 RETURN ON ASSETS (ROA)

Definition: Profitability metric measuring efficiency of asset utilization

Formula:

```
ROA = Net Income / Total Assets × 100%

ROA = Net Income / Average Total Assets × 100%
```

Interpretation:

- ROA > 10%: Excellent asset efficiency
- ROA 5-10%: Good
- ROA 2-5%: Average
- ROA < 2%: Below average

2.8 PROFIT MARGINS

Definition: Series of profitability metrics at different levels

Formulas:

```
Gross Profit Margin = (Revenue - COGS) / Revenue × 100%
Where COGS = Cost of Goods Sold

Operating Profit Margin = Operating Income / Revenue × 100%

Net Profit Margin = Net Income / Revenue × 100%

EBITDA Margin = EBITDA / Revenue × 100%
Where EBITDA = Earnings Before Interest, Taxes, Depreciation, Amortization
```

Interpretation:

- Gross Margin: Production efficiency (industry dependent)
- Operating Margin: Operational efficiency (excluding financing)
- Net Margin: Overall profitability (after all expenses)
- Trend: Improving or declining margins indicate operational changes

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PART 3: MACHINE LEARNING MODEL METRICS

3.1 MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

Definition: Percentage error metric; measures average magnitude of errors without direction

Formula:

```
MAPE = (100 / n) × Σ|((Actual_i - Predicted_i) / Actual_i)|

Where:
n = number of predictions
Actual_i = true value at time i
Predicted_i = model prediction at time i
|...| = absolute value
```

Scale:

- MAPE < 3%: Excellent accuracy
- MAPE 3-5%: Good accuracy
- MAPE 5-10%: Acceptable accuracy
- MAPE 10-15%: Fair accuracy
- MAPE > 15%: Poor accuracy

Advantages:

- Scale-independent (percentage-based)
- Easy to interpret
- Suitable for comparing across different scales

Disadvantages:

- Undefined when actual value = 0
- Can be misleading with small values
- Biased toward underestimation

3.2 ROOT MEAN SQUARED ERROR (RMSE)

Definition: Measures average magnitude of prediction errors; penalizes large errors heavily

Formula:

```
RMSE = √[(1/n) × Σ(i=1 to n) (Actual_i - Predicted_i)²]

Where:
n = number of predictions
Σ = summation over all predictions
(...)² = squared difference
√ = square root
```

Scale:

- Same units as target variable (e.g., ₹ for stock prices)
- RMSE < 2% of average price: Good (for stocks)
- RMSE 2-5% of average price: Acceptable
- RMSE > 5% of average price: Poor

Advantages:

- Penalizes large errors more (better for critical predictions)
- Differentiable (suitable for optimization)
- Widely used in regression

Disadvantages:

- Scale-dependent (hard to compare across datasets)
- Heavily influenced by outliers

3.3 MEAN ABSOLUTE ERROR (MAE)

Definition: Average absolute error; measures average magnitude without direction

Formula:

```
MAE = (1/n) × Σ|Actual_i - Predicted_i|

Where:
n = number of predictions
|...| = absolute value
```

Advantages:

- Robust to outliers (linear scale)
- Same units as target variable
- Intuitive interpretation

Disadvantages:

- Doesn't penalize large errors as heavily as RMSE
- Not differentiable at zero (optimization challenge)

3.4 R-SQUARED (COEFFICIENT OF DETERMINATION)

Definition: Proportion of variance in dependent variable explained by independent variable(s)

Formula:

```
R² = 1 - (SS_res / SS_tot)

Where:
SS_res = Σ(Actual_i - Predicted_i)² (residual sum of squares)
SS_tot = Σ(Actual_i - Mean_Actual)² (total sum of squares)

Range: 0 to 1 (can be negative for very poor models)
```

Interpretation:

- R^2 = 1.0: Perfect prediction
- R^2 = 0.9-0.99: Excellent fit (90-99% variance explained)
- R^2 = 0.7-0.9: Good fit (70-90% variance explained)
- R^2 = 0.5-0.7: Acceptable fit (50-70% variance explained)
- R^2 = 0-0.5: Poor fit (< 50% variance explained)
- R^2 < 0: Model worse than simple mean prediction

3.5 MEAN SQUARED ERROR (MSE)

Definition: Average squared error; baseline for RMSE calculation

Formula:

```
MSE = (1/n) × Σ(i=1 to n) (Actual_i - Predicted_i)²

Note: RMSE = √MSE
```

Advantages:

- Penalizes large errors heavily
- Differentiable and convex (optimization friendly)

Disadvantages:

- Same units as (target variable)^2 - not intuitive
- Heavily influenced by outliers

3.6 PREDICTION INTERVAL & CONFIDENCE INTERVAL

Definition: Range around prediction accounting for uncertainty

Formula:

```
Prediction Interval = Point_Estimate ± (Z_critical × Standard_Error)

95% Prediction Interval = Prediction ± (1.96 × σ_error)

Where:
Z_critical = critical value from standard normal distribution
Standard_Error = estimated standard deviation of error
1.96 = z-score for 95% confidence level
```

Interpretation:

- Wider interval: Higher uncertainty
- Narrower interval: Higher confidence
- 95% interval: True value expected to fall within range 95% of time

---

PART 4: VOLATILITY & RISK METRICS

4.1 STANDARD DEVIATION (σ)

Definition: Measures dispersion of price movements around mean; volatility indicator

Formula:

```
σ = √[Σ(Price_i - Mean_Price)² / (n - 1)]

Where:
n = number of observations
Mean_Price = average price over period
(n-1) = degrees of freedom (sample std dev)

Historical Volatility = σ of daily returns over period
Realized Volatility = √[Σ(Daily_Return_i)² / n]
```

Interpretation:

- σ < 15%: Low volatility, stable stock
- σ 15-30%: Moderate volatility, typical stock
- σ 30-50%: High volatility, risky stock
- σ > 50%: Very high volatility, speculative

4.2 BETA (β)

Definition: Systematic risk measure; volatility relative to market index

Formula:

```
β = Covariance(Stock_Return, Market_Return) / Variance(Market_Return)

β = Correlation(Stock, Market) × (σ_Stock / σ_Market)

Also: Stock_Return = α + β × Market_Return + ε
Where α = alpha (excess return)
      ε = error term
```

Interpretation:

- β = 1.0: Stock moves with market
- β > 1.0: More volatile than market (amplified moves)
- β < 1.0: Less volatile than market (dampened moves)
- β < 0: Inverse relationship with market (rare)

Examples:

- β = 1.5: If market up 10%, stock expected up 15%
- β = 0.8: If market down 10%, stock expected down 8%

4.3 SHARPE RATIO

Definition: Risk-adjusted return metric; reward per unit of risk taken

Formula:

```
Sharpe Ratio = (Portfolio_Return - Risk_Free_Rate) / σ_Portfolio

Where:
Portfolio_Return = average return of strategy/prediction
Risk_Free_Rate = return on safe asset (e.g., 5% annual for India)
σ_Portfolio = standard deviation of returns
```

Interpretation:

- SR > 2.0: Excellent risk-adjusted returns
- SR 1.0-2.0: Good risk-adjusted returns
- SR 0.5-1.0: Acceptable risk-adjusted returns
- SR < 0.5: Poor risk-adjusted returns (not worth risk)
- Negative SR: Negative returns (avoid)

Target for Stock Prediction System: SR > 1.5

4.4 MAXIMUM DRAWDOWN (MDD)

Definition: Worst peak-to-trough decline; maximum loss from peak to bottom

Formula:

```
Running_Maximum_t = max(Cumulative_Return_i) for i ≤ t

Drawdown_t = (Running_Maximum_t - Cumulative_Return_t) / Running_Maximum_t

Maximum_Drawdown = max(Drawdown_t) for all t

Cumulative_Return = (Final_Value - Initial_Value) / Initial_Value × 100%
```

Interpretation:

- MDD < 10%: Low drawdown risk
- MDD 10-20%: Moderate drawdown risk
- MDD 20-30%: High drawdown risk
- MDD > 30%: Very high drawdown risk

Context: If system predicts +8% annual return with MDD -15%, could lose 15% before recovering

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4.5 VALUE AT RISK (VaR)

Definition: Maximum loss expected at given confidence level over period

Formula:

```
VaR_95% = Portfolio_Value × Z_95% × σ_daily_return × √days

Where:
Z_95% = 1.65 (z-score for 95% confidence)
σ_daily_return = standard deviation of daily returns
√days = square root of number of days

Example:
If portfolio = ₹1,00,000, daily σ = 2%, 95% VaR for 1 day:
VaR = 1,00,000 × 1.65 × 0.02 × 1 = ₹3,300
(95% confident loss won't exceed ₹3,300 tomorrow)
```

Interpretation:

- VaR 95%: 95% confident loss won't exceed this amount
- VaR 99%: 99% confident loss won't exceed this amount
- Used for position sizing and risk limits

PART 5: ENSEMBLE MODEL PARAMETERS

5.1 WEIGHTED ENSEMBLE PREDICTION

Definition: Combining predictions from multiple models using weights

Formula:

```
Ensemble_Prediction = Σ(Weight_i × Model_Prediction_i)

Where:
Weight_i = importance weight for model i
Σ(Weight_i) = 1.0 (constraint: weights sum to 1)

Inverse MSE Weighting:
Weight_i = (1/MSE_i) / Σ(1/MSE_j)

Example:
If MSE_LSTM = 100, MSE_XGBoost = 150, MSE_RF = 200
Inverse_LSTM = 0.01, Inverse_XGBoost ≈ 0.0067, Inverse_RF = 0.005
Sum = 0.0217
Weight_LSTM = 0.01/0.0217 = 0.461 (46.1%)
Weight_XGBoost = 0.0067/0.0217 = 0.309 (30.9%)
Weight_RF = 0.005/0.0217 = 0.230 (23.0%)

Ensemble_Price = 0.461×1262.50 + 0.309×1268.00 + 0.230×1270.00
             = ₹1,266.89
```

Ensemble Confidence:

```
Ensemble_Confidence = Σ(Weight_i × Confidence_i)

Where:
Confidence_i = 1 - (MAPE_i / 100)

Example:
If MAPE_LSTM = 4.2%, MAPE_XGBoost = 3.8%, MAPE_RF = 5.1%
Conf_LSTM = 1 - 0.042 = 0.958
Conf_XGBoost = 1 - 0.038 = 0.962
Conf_RF = 1 - 0.051 = 0.949

Ensemble_Conf = 0.461×0.958 + 0.309×0.962 + 0.230×0.949 = 0.957 (95.7%)
```

PART 6: MULTI-AGENT FUSION PARAMETERS

6.1 TECHNICAL INDICATOR COMPOSITE SCORE

Definition: Weighted average of all technical indicator signals

Formula:

```
Technical_Score = (w_RSI × RSI_Score) +
                 (w_MACD × MACD_Score) +
                 (w_BB × BB_Score) +
                 (w_Vol × Vol_Score)

Standard Weights:
w_RSI = 0.25
w_MACD = 0.35   (trend indicator, weighted higher)
w_BB = 0.20
w_Vol = 0.20

Σ(weights) = 1.0

Individual Scores Range: [-1.0, +1.0]
Technical_Score Range: [-1.0, +1.0]

Individual Signal Mappings:
RSI Score:
  RSI > 70: +0.8 (overbought)
  RSI 60-70: +0.4 (moderately overbought)
  RSI 40-60: 0.0 (neutral)
  RSI 30-40: -0.4 (moderately oversold)
  RSI < 30: -0.8 (oversold)

MACD Score:
  MACD > Signal, Hist > 0: +0.9 (strong bullish)
  MACD > Signal, Hist ≤ 0: +0.3 (weak bullish)
  MACD < Signal, Hist < 0: -0.9 (strong bearish)
  MACD < Signal, Hist ≥ 0: -0.3 (weak bearish)

Bollinger Bands Score (%B Position):
  %B > 0.9: +0.7 (at upper band, overbought)
  %B 0.6-0.9: +0.3 (upper half, bullish)
  %B 0.4-0.6: 0.0 (middle, neutral)
```

```
  %B 0.1-0.4: -0.3 (lower half, bearish)
  %B < 0.1: -0.7 (at lower band, oversold)

Volume Score:
  OBV > OBV_SMA, Price > Price_SMA: +0.6 (strong confirmation)
  OBV > OBV_SMA, Price < Price_SMA: -0.4 (divergence warning)
  OBV < OBV_SMA, Price > Price_SMA: +0.3 (weak bullish)
  OBV < OBV_SMA, Price < Price_SMA: -0.7 (strong bearish)
```

6.2 SENTIMENT COMPOSITE SCORE

Definition: Weighted fusion of news, social, and fundamental sentiment

Formula:

```
Composite_Sentiment = (w_news × Score_news) +
                      (w_social × Score_social) +
                      (w_fundamental × Score_fundamental)

Weights (by reliability):
w_news = 0.45          (Financial news most reliable)
w_social = 0.25        (Twitter/social media moderate)
w_fundamental = 0.30   (Company fundamentals high)

Σ(weights) = 1.0

Range: [-1.0, +1.0]

Individual Sentiment Scores:
News Sentiment (using VADER or DistilBERT):
  Score = Positive_prob - Negative_prob
  Range: [-1.0, +1.0]

Social Sentiment:
  Tweet_Sentiment = Σ(sentiment_i) / n_tweets
  Range: [-1.0, +1.0]
  -1.0 = all negative, 0.0 = neutral, +1.0 = all positive

Fundamental Sentiment:
  EPS_Growth = (EPS_current - EPS_previous) / EPS_previous
  If EPS_Growth > 15%: +0.8
  If EPS_Growth 0-15%: +0.4
  If EPS_Growth = 0%: 0.0
  If EPS_Growth < 0%: -0.6
```

6.3 MULTI-AGENT CONSENSUS SCORE

Definition: Weighted consensus from 4 specialized agents

Formula:

```
Agent_Consensus = (w_news × News_Agent_Score) +
                  (w_fundamental × Fundamental_Agent_Score) +
                  (w_pattern × Pattern_Agent_Score) +
                  (w_risk × Risk_Agent_Score)

Weights:
w_news = 0.25          (Sentiment evaluation)
w_fundamental = 0.30   (Company health)
w_pattern = 0.25       (Chart patterns)
w_risk = 0.20          (Risk factors)

Σ(weights) = 1.0

Agent_Consensus Range: [-1.0, +1.0]

Agent Agreement Metric:
Agent_Agreement = 1.0 - (Σ(|Score_i - Consensus|) / (4 × 2.0))

Where:
Score_i = individual agent score
Consensus = Agent_Consensus
(4 × 2.0) = 4 agents × max range of 2.0

Agent_Agreement Range: [0.0, 1.0]
  1.0 = perfect agreement (all agents aligned)
  0.5 = moderate agreement
  0.0 = total disagreement (agents conflict)
```

6.4 FINAL PREDICTION FUSION

Definition: Complete fusion formula combining ensemble, technical, and sentiment

Formula:

```
Step 1: Price Change Prediction
Ensemble_Pred = Weighted average of 3 models
Predicted_Change_Pct = ((Ensemble_Pred - Current_Price) / Current_Price) × 100

Step 2: Technical Adjustment
Adjusted_Change = Predicted_Change_Pct + (Technical_Score × 2.0)

Example:
If Pred_Change = +2.5%, Technical_Score = +0.3
Adjusted_Change = 2.5 + (0.3 × 2.0) = 3.1%

Step 3: Sentiment Adjustment
Final_Change = Adjusted_Change + (Composite_Sentiment × 1.5)

Example:
If Adjusted = +3.1%, Sentiment = +0.5
Final_Change = 3.1 + (0.5 × 1.5) = 3.85%

Step 4: Final Price
Final_Predicted_Price = Current_Price × (1 + Final_Change/100)

Example:
If Current = ₹1,220, Final_Change = +3.85%
Final_Price = 1,220 × 1.0385 = ₹1,267.00
```

6.5 FINAL CONFIDENCE SCORE

Definition: Combined confidence from all three sources

Formula:

```
Base_Confidence = Ensemble_Confidence (from MAPE)
                = 1 - (Ensemble_MAPE / 100)
Range: [0.0, 1.0]

Technical_Confidence = |Technical_Score|
                     = absolute value of technical score
Range: [0.0, 1.0]
(Stronger technical signal = higher confidence)

Agent_Confidence = Agent_Agreement
Range: [0.0, 1.0]
(Higher agreement among agents = higher confidence)

Final_Confidence = (0.50 × Base_Confidence) +
                   (0.25 × Technical_Confidence) +
                   (0.25 × Agent_Confidence)

Convert to Percentage: Final_Confidence_Pct = Final_Confidence × 100

Example:
Base = 0.87, Technical = 0.65, Agent_Agreement = 0.78
Final = (0.50 × 0.87) + (0.25 × 0.65) + (0.25 × 0.78)
      = 0.435 + 0.1625 + 0.195
      = 0.7925 → 79.25%
```

6.6 TRADING RECOMMENDATION SCORE

Definition: Combined score determining buy/sell/hold recommendation

Formula:

```
Combined_Score = (0.40 × Technical_Score) +
                 (0.35 × Agent_Consensus) +
                 (0.25 × (Ensemble_Price_Change / 5.0))

Normalization:
Ensemble_Price_Change / 5.0 brings percentage to [-1, +1] range
(If price change +5%, score = +1; if -5%, score = -1)

Combined_Score Range: [-1.0, +1.0]

Recommendation Rules:
If Combined_Score > +0.50: BUY (Strong buy signal)
If Combined_Score > +0.20 AND ≤ +0.50: BUY_WEAK (Mild buy)
If Combined_Score ≥ -0.20 AND ≤ +0.20: HOLD (Neutral)
If Combined_Score < -0.20 AND ≥ -0.50: SELL_WEAK (Mild sell)
If Combined_Score < -0.50: SELL (Strong sell signal)
```

6.7 RISK LEVEL ASSESSMENT

Definition: Overall risk classification based on multiple factors

Formula:

```
Risk_Agent_Score = -(Volatility_normalized + Beta_normalized) / 2

Where:
Volatility_normalized = (Volatility - Min_Volatility) / (Max_Volatility - Min_Volatility)
Beta_normalized = (Beta - 0.5) / 2 (assuming beta 0.5-2.5 range)

Risk_Level Assignment:
If Risk_Agent_Score < -0.6 OR Volatility > 35%:
  Risk_Level = HIGH
  Interpretation: High risk, large potential moves, wider stop-losses

If Risk_Agent_Score < -0.3 OR Volatility 25-35%:
  Risk_Level = MEDIUM
  Interpretation: Moderate risk, typical stock, normal stops

If Risk_Agent_Score ≥ -0.3 OR Volatility < 25%:
  Risk_Level = LOW
  Interpretation: Lower risk, stable stock, tighter stops
```

PART 7: KEY TARGETS & THRESHOLDS

7.1 MODEL PERFORMANCE TARGETS

```
LSTM Model:
- MAPE Target: < 3.5%
- RMSE Target: < 2% of average stock price
- Lookback Window: 60 trading days
- Architecture: LSTM64 → LSTM32 → Dense16 → Dense1

Ensemble Model:
- Ensemble MAPE: < 3.0%
- Individual Model MAPE: 3.5-5.5%
- Weight Rebalancing: Weekly
- Backtesting Accuracy: 65-75% directional accuracy

Risk Metrics:
- Sharpe Ratio Target: > 1.5
- Maximum Drawdown: < 15%
- Value at Risk (95%): < 5% daily loss
```

7.2 TECHNICAL INDICATOR THRESHOLDS

```
RSI:
- Overbought: > 70
- Oversold: < 30
- Neutral: 40-60
- Signal Generation: ±0.8 (extreme), ±0.4 (moderate)

MACD:
- Bullish: MACD > Signal + Histogram > 0
- Bearish: MACD < Signal + Histogram < 0
- Signal Generation: ±0.9 (strong), ±0.3 (weak)

Bollinger Bands:
- Upper Band Alert: Price > SMA + 2σ (overbought)
- Lower Band Alert: Price < SMA - 2σ (oversold)
- %B Threshold: 0.1 (lower) to 0.9 (upper)

Volume:
- Confirmation: OBV moving with price
- Divergence: OBV opposite to price direction
- Strength: OBV > 20-day average
```

7.3 FUNDAMENTAL METRIC RANGES

```
P/E Ratio:
- Undervalued: < 15
- Fair Value: 15-25
- Overvalued: > 25
- Context: Compare to industry, historical average

P/B Ratio:
- Low: < 1.0 (value opportunity)
- Fair: 1.0-3.0 (normal range)
- High: > 3.0 (expensive or high-growth)

Dividend Yield:
- High: > 5% (verify sustainability)
- Moderate: 2-5% (typical)
- Low: < 2% (growth focus)

D/E Ratio:
- Conservative: < 0.5
- Moderate: 0.5-1.5
- High Risk: > 2.0

ROE:
- Excellent: > 15%
- Good: 10-15%
- Average: 5-10%
- Poor: < 5%
```

PART 8: DATA QUALITY STANDARDS

8.1 INPUT DATA REQUIREMENTS

```
Stock Price Data:
- Minimum History: 5 years (1,200+ trading days)
- Minimum Daily Volume: ₹10 Crore
- Data Frequency: 1-minute to daily
- Missing Data: < 5% allowed
- Outlier Threshold: Price changes > 10% in single day (flag for validation)

News & Sentiment Data:
- Real-time: Updated within 30 minutes of news
- Historical: 12 months minimum
- Sources: Reuters, Bloomberg, company filings, Twitter
- Validation: Cross-check multiple sources

Fundamental Data:
- Quarterly: Latest 4 quarters (for TTM calculations)
- Annual: Latest 5 years
- Update Frequency: When company reports
- Validation: Match with official filings
```

8.2 DATA VALIDATION CHECKS

```
Price Data:
- Check for gaps (missing trading days)
- Verify OHLC relationship (High ≥ Low, Close within range)
- Detect extreme outliers (> 10% single day)
- Validate volume (> 0 on trading days)

Volume Data:
- Check for trading day consistency
- Flag unusually high/low volume
- Verify no negative volumes

Technical Indicators:
- Verify calculation logic
- Check for NaN/Inf values
- Validate indicator ranges
  * RSI: 0-100
  * MACD: unrestricted (difference of EMAs)
  * Bollinger Bands: prices within bands
  * OBV: cumulative value
```

PART 9: REAL-WORLD IMPLEMENTATION NOTES

9.1 TRADING SIMULATION PARAMETERS

```
Backtesting Period:
- Minimum: 2 years historical data
- Walk-Forward Validation: 252 trading days (1 year) test sets
- Retraining: Monthly with new data

Transaction Costs:
- Brokerage: 0.1% (per trade)
- STT (Security Transaction Tax): 0.1% (buy) + 0.1% (sell)
- Total Round-Trip Cost: 0.3-0.4%
- Slippage: 0.05-0.2% (market impact)

Profit Requirement:
- To break even after costs: +0.4%
- Minimum profitable trade: +0.5-1.0%
- Target for system: +3-5% average per prediction

Portfolio Sizing:
- Position Size = Portfolio × Risk_Percentage / (Stop_Loss_in_%)
- Risk per Trade: 1-2% of portfolio max
- Diversification: Min 5-10 stocks
- Concentration Limit: Max 20% in single stock
```

9.2 PERFORMANCE BENCHMARKING

```
Comparison Benchmarks:
- Nifty 50 Index: Target to beat this
- Sector Average: Compare within sector
- Risk-Free Rate: 5% annual (India)
- Active Management: +3-5% alpha expectation

Backtesting Results Expected:
- Directional Accuracy: 65-75% (better than random)
- Winning Trades: 55-60%
- Average Win: +2-3%
- Average Loss: -1-2%
- Win/Loss Ratio: 1.5-2.0

Real Trading Results Expected:
- Live accuracy: 5-10% lower than backtest
- Slippage: 0.05-0.2% impact per trade
- Liquidity: May not fill at exact predicted price
- Drawdown: Peak-to-trough -10 to -20%
```

9.3 ADJUSTMENT FACTORS

```
Market Condition Adjustments:
- Trending Market (ADX > 25):
  * Increase weight on MACD (trend indicator)
  * Technical_Score multiplier: 1.2x

- Range-Bound Market (ADX < 20):
  * Increase weight on RSI/Bollinger Bands (mean reversion)
  * Technical_Score multiplier: 0.8x

- High Volatility (ATR > 2%):
  * Increase stop-loss distance
  * Reduce position size
  * Increase confidence threshold to 80%+

- Low Liquidity (Volume < ₹1 Cr daily):
  * Avoid predictions
  * Increase slippage assumption
  * Reduce position size to 50% normal

Time-Based Adjustments:
- Pre-Market (9:00-9:15): No trading (illiquid)
- Market Open (9:15-10:00): Higher volatility, use wider stops
- Lunch Time (12:00-1:00): Lower volume, wider spreads
- Market Close (3:15-3:30): Volume spike, use market orders
- Earnings Announcements: Avoid or use very wide stops
```

SUMMARY TABLE: KEY FORMULAS AT A GLANCE

Metric	Formula	Key Parameter	Target/Interpretati
RSI	100 - 100/(1+RS)	Period: 14	>70 overbought, <30 oversold
MACD	EMA12 - EMA26, Signal=EMA9	12-26-9	>Signal = bullish

Metric	Formula	Key Parameter	Target/Interpretati
Bollinger	SMA ± 2σ	20-period, 2σ	Price outside bands extreme
OBV	Cumulative vol. based on close	20-day MA	Rising with price = bullish
EPS	(Net Income - Pref Div) / Shares	Quarterly/Annual	Increasing = better profitability
P/E	Stock Price / EPS	Trailing/Forward	<15 undervalued, >2 expensive
MAPE	100/n × Σ|error/actual|	n = # predictions	<3% excellent, <5% good
RMSE	$\sqrt{[\Sigma(\text{error})^2/n]}$	n = # predictions	<2% of avg price = good
R^2	1 - (SS_res/SS_tot)	Variance explained	>0.7 = good fit
Beta	Cov(stock, market) / Var(market)	14-252 days	>1 more volatile, <1 stable
Sharpe	(Return - Risk_Free) / σ_return	Risk_free = 5%	>1.5 = good risk-adjusted
Volatility	$\sqrt{[\Sigma(\text{return - avg})^2/(n-1)]}$	20-30 days	<15% low, >30% high
Technical_Score	0.25RSI + 0.35MACD + 0.20BB + 0.20OBV	Weights	[-1, +1] range
Sentiment	0.45news + 0.25social + 0.30fund	Weights	[-1, +1] range

Metric	Formula	Key Parameter	Target/Interpretati
Agent_Consensus	0.25news + 0.30fund + 0.25pattern + 0.20risk	Weights	[-1, +1] range
Final_Price	Current × (1 + Final_Change/100)	Change = 3.85%	Predicted target pri
Final_Confidence	0.50base + 0.25tech + 0.25agent	% score	>75% good, >85% excellent
Recommendation	0.40tech + 0.35agent + 0.25price_change	Combined	>0.5 BUY, <-0.5 SEI

DISCLAIMER

This document provides technical definitions and formulas for stock market analysis and machine learning model evaluation. All formulas are mathematically correct and industry-standard, but:

1. **Past performance ≠ Future Results**: Historical accuracy doesn't guarantee future predictions
2. **Market Efficiency**: Some stocks are semi-efficient; predictions are probabilistic
3. **Not Financial Advice**: Always consult professional advisors before trading
4. **Risk Acknowledgment**: Stock market investments carry inherent risks of loss
5. **System Limitations**: Even well-built systems can't consistently beat the market long-term

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Document Version: 1.0 **Last Updated**: December 9, 2025 **Total Formulas Documented**: 65+ **Coverage**: Technical Analysis, Fundamentals, ML Metrics, Risk Metrics, Fusion Logic